

MATH0071 Spectral Theory

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0071
<i>Level:</i>	7 (UG), 7 (PG)
<i>Normal student group(s):</i>	UG: Y4 Mathematics programmes PG: MSc Mathematics
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	90% examination, 10% coursework
<i>Normal Pre-requisites:</i>	MATH0018 (Functional Analysis)
<i>Lecturer:</i>	Prof L Parnowski

Course Description and Objectives

Spectral theory came to prominence when quantum mechanics was introduced in modern physics. In quantum mechanics classical quantities (position, momentum etc) are represented by operators (bounded, unbounded, self-adjoint etc). The eigenvalues of these operators are the only precise measurements of the quantity. Without requiring knowledge of physics, this course introduces the fundamentals of such operators and the space of their eigenvalues, which is called the spectrum.

Recommended Texts

- EB Davies, *Linear Operators and Their Spectra* (Cambridge)
- W Arveson, *A short course on spectral theory* (Springer)

Detailed Syllabus

Banach and Hilbert Spaces. Orthogonal projections. Orthonormal bases. Fourier expansion. Riesz representation theorem. Linear operators (bounded and unbounded). Adjoint operator. Symmetric, normal, self-adjoint and compact operators. Resolvent. Spectra of linear operators: classification and properties. Spectral theorem for compact self-adjoint operators. Applications to differential and integral equations.

Further topics are chosen from: spectral theorem for bounded self-adjoint operators; unbounded operators and applications; Fredholm operators; extensions of symmetric operators; quadratic forms.